

ROVANIEMI RAUTATIEASEMA, Finland

WMO: 028470

Lat: 66.498N

Lon: 25.708E

Elev: 270

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-13.2	-24.1	1.5	-18.5	-18.1	2.1	-13.1	17.9	27.4	15.8	27.4	1.7	60	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	79.4	62.5	75.6	60.9	72.1	59.5	65.9	73.7	63.5	72.1	61.5	69.1	6.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	86.7	69.7	60.5	79.5	65.8	58.4	73.6	64.3	30.8	73.1	29.0	72.2	27.6	68.7	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.5	12.9	11.4	DB	-22.8	82.1	6.9	4.4	-27.8	85.2	-31.8	87.8	-35.7	90.3	-40.7	93.5	
			WB	-22.8	68.6	6.6	2.9	-27.5	70.7	-31.3	72.4	-35.0	74.0	-39.8	76.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	11.0	14.4	22.8	34.0	46.7	55.0	62.0	57.2	48.5	35.1	25.9	18.6	
	DBStd	19.10	11.96	13.61	8.92	6.66	8.28	7.04	6.39	6.07	5.99	6.89	9.26	11.72	
	HDD50	5992	1210	996	844	481	165	29	1	11	98	463	722	972	
	HDD65	10629	1675	1416	1309	930	570	309	135	252	496	928	1172	1437	
	CDD50	901	0	0	0	1	62	179	373	234	52	0	0	0	
	CDD65	62	0	0	0	0	2	9	41	10	0	0	0	0	
	CDH74	495	0	0	0	0	38	64	325	68	0	0	0	0	
	CDH80	89	0	0	0	0	3	7	68	11	0	0	0	0	
Wind	WSAvg	5.2	4.3	5.0	5.5	5.2	6.0	5.7	5.2	5.2	4.9	4.9	4.9	5.0	
Precipitation	PrecAvg	23.9	1.7	1.4	1.3	1.3	1.7	2.4	3.0	2.6	2.3	2.1	2.1	1.9	
	PrecMax	32.4	2.8	2.9	2.7	3.3	3.9	4.3	5.2	5.9	4.6	3.6	3.7	3.8	
	PrecMin	16.9	0.6	0.2	0.4	0.4	0.3	0.5	0.8	0.4	0.5	0.6	0.6	0.6	
	PrecStd	4.3	0.6	0.7	0.6	0.7	0.8	1.1	1.2	1.4	1.1	0.8	0.8	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.5	39.2	43.6	61.9	78.7	81.1	86.4	81.6	66.4	51.9	43.0	37.6	
		MCWB	33.5	36.2	36.6	47.9	57.6	63.5	67.2	67.5	56.8	48.2	42.3	36.3	
	2%	DB	32.2	35.2	39.7	52.3	73.2	75.6	81.5	74.9	63.1	49.2	40.6	35.2	
		MCWB	31.8	33.6	34.3	41.8	57.2	60.3	63.3	61.4	54.9	46.1	39.7	34.2	
	5%	DB	30.4	33.5	36.8	47.9	66.5	71.2	77.4	70.9	60.5	47.0	38.4	33.9	
		MCWB	29.9	32.5	32.8	39.6	52.9	57.1	62.1	60.1	54.1	44.8	37.7	33.3	
	10%	DB	27.9	31.6	34.5	44.4	61.3	67.3	73.9	67.3	57.7	44.8	36.4	32.8	
		MCWB	27.3	30.7	32.1	37.8	49.9	55.1	61.0	58.7	53.3	43.1	35.8	32.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.7	36.4	37.6	48.0	59.7	65.7	69.6	68.0	59.5	50.0	42.3	36.7
			MCDB	34.1	38.5	42.1	61.0	72.8	76.4	77.6	77.0	63.4	50.9	42.6	37.6
2%		WB	31.9	34.1	35.1	43.5	57.1	62.8	66.9	64.5	56.8	46.6	39.6	34.5	
		MCDB	32.2	34.9	38.1	50.3	69.9	73.4	74.2	71.8	60.6	48.0	40.1	35.0	
5%		WB	30.0	32.6	33.8	40.4	54.4	60.2	64.7	61.9	55.5	45.0	37.9	33.4	
		MCDB	30.3	33.4	35.6	46.6	65.7	68.0	73.5	67.5	59.0	46.9	38.4	33.8	
10%		WB	27.2	30.8	32.6	38.4	51.2	57.2	62.6	60.0	53.8	43.1	35.9	32.2	
		MCDB	27.8	31.6	34.3	43.3	59.3	63.1	71.1	64.8	56.9	44.7	36.3	32.7	
Mean Daily Temperature Range	5% DB	MDBR	12.6	12.5	16.0	15.4	16.4	14.8	15.5	14.2	12.2	9.0	8.4	10.8	
		MCDBR	13.9	9.9	14.9	21.0	25.3	20.5	21.1	20.0	16.8	11.5	7.9	8.2	
		MCWBR	14.0	8.8	11.0	12.3	11.6	8.9	7.7	8.7	9.8	9.0	7.5	8.1	
	5% WB	MCDBR	14.2	9.5	10.8	18.7	23.0	16.5	17.7	15.7	12.8	10.1	7.5	8.5	
		MCWBR	14.4	8.5	8.6	11.5	11.5	9.0	8.2	8.3	8.4	8.6	7.3	8.4	
Clear-Sky Solar Irradiance	taub	0.199	0.267	0.299	0.334	0.345	0.346	0.360	0.343	0.319	0.291	0.347	N/A		
	taud	1.935	2.103	2.166	2.242	2.445	2.469	2.448	2.479	2.532	2.346	2.640	N/A		
	Ebn at Noon	64	180	238	260	272	274	265	258	239	177	68	N/A		
	Edh at Noon	5	17	27	33	30	30	30	26	20	14	5	N/A		
All-Sky Solar Radiation	RadAvg	22	174	623	1151	1495	1647	1548	1124	592	207	42	2		
	RadStd	2	31	65	100	137	125	147	100	64	41	5	0		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (42 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page